D1.2 Protein synthesis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d1-2.html

Written for this site from the course textbook and class slides, with page references for each statement. The chapter is on pages 615–624 of the course textbook. Image credits are on the last slide of the slides for this subtopic.

Transcription

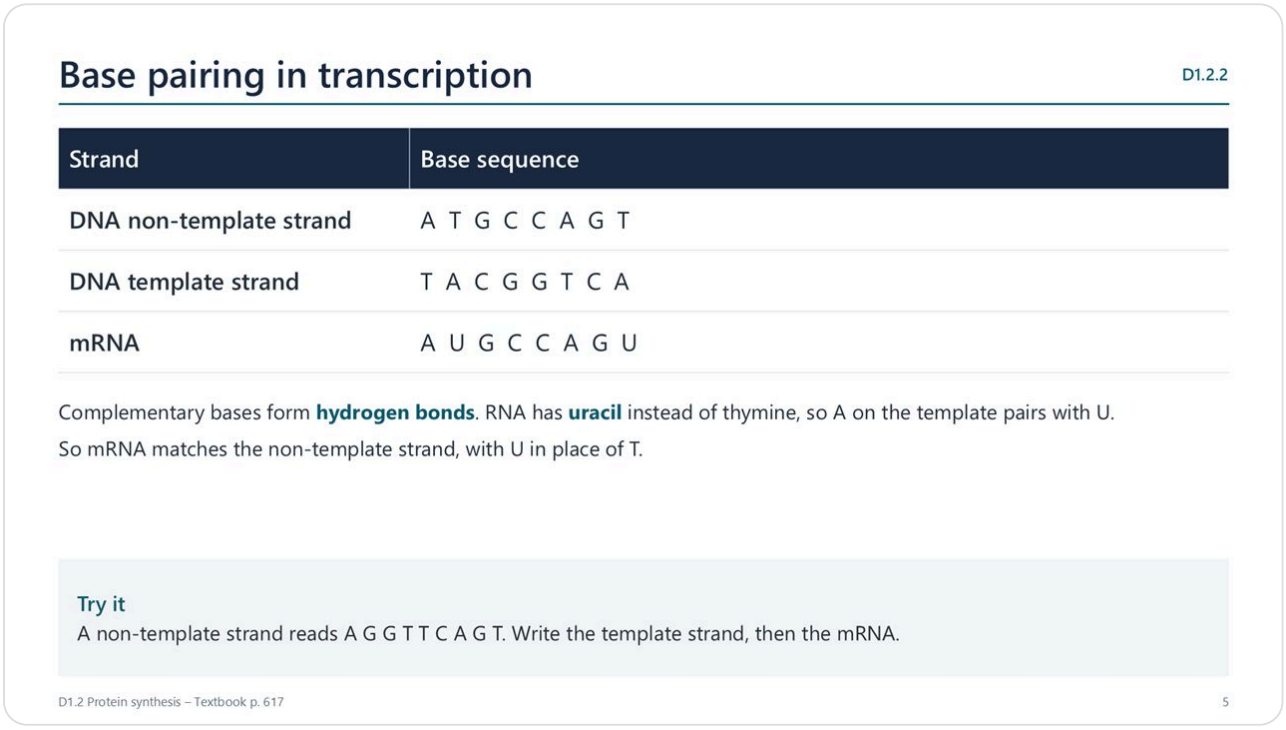
D1.2.1 Transcription: making RNA from a DNA template

p. 616

RNA polymerase binds to DNA at the start of a gene and separates the strands as it moves along. One strand acts as the template. RNA nucleotides pair with it, and RNA polymerase links them with covalent bonds. The RNA then separates, and the double helix reforms.

D1.2.2 Hydrogen bonds and base pairing

p. 617



Base pairing in transcription

D1.2.2

Strand	Base sequence
DNA non-template strand	A T G C C A G T
DNA template strand	T A C G G T C A
mRNA	A U G C C A G U

Complementary bases form **hydrogen bonds**. RNA has **uracil** instead of thymine, so A on the template pairs with U.
So mRNA matches the non-template strand, with U in place of T.

Try it
A non-template strand reads A G G T T C A G T. Write the template strand, then the mRNA.

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Hydrogen bonds form between complementary bases. Adenine on the template pairs with **uracil** in RNA, as RNA has no thymine. The mRNA therefore matches the non-template strand, with U in place of T.

D1.2.3 Why DNA templates stay stable

p. 618

Transcription copies a base sequence without changing the DNA, so a gene can be transcribed many times. This matters in non-dividing body cells, such as neurons, where the same sequences must be conserved throughout life.

D1.2.4 Transcription and gene expression

p. 618

Not all genes in a cell are expressed at any given time. Transcription is the first stage of expression, so it is a key point for switching a gene on or off. Liver cells and neurons have the same genes, but they transcribe different ones.

Translation

D1.2.5 Translation: making polypeptides from mRNA

p. 618

The base sequence of mRNA decides the amino acid sequence of a polypeptide. The language changes here from bases to amino acids, which is why this stage is called translation.

D1.2.6 What mRNA, ribosomes and tRNA do

p. 619

Molecule	Role
mRNA	Carries codons from the gene, and binds to the small subunit of a ribosome
Ribosome	Holds mRNA and tRNA, with two tRNAs bound to the large subunit at once; forms peptide bonds
tRNA	Carries a specific amino acid, and pairs with a codon using its anticodon

D1.2.7 Codons and anticodons

p. 619

A **codon** is three bases on mRNA, and an **anticodon** is three complementary bases on tRNA. Hydrogen bonds hold them together. For example, codon G C A pairs with anticodon C G U.

D1.2.8 Features of the genetic code

p. 620

- **Triplet:** two bases give only 16 combinations, too few for 20 amino acids. Three bases give 64.
- **Degenerate:** more than one codon can code for the same amino acid.
- **Universal:** almost all organisms read codons in the same way, which suggests a common ancestor.

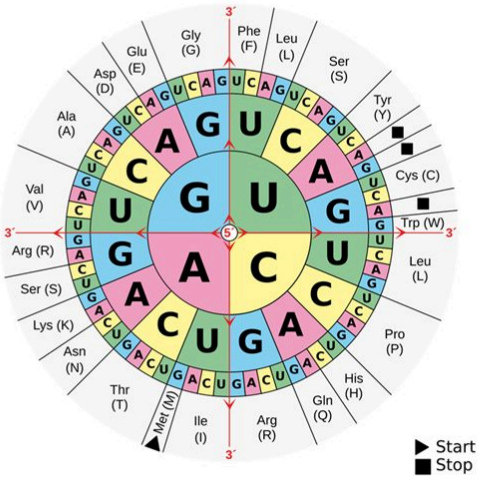
Watch out: degenerate does not mean that one codon can give several amino acids.

D1.2.9 Reading a table of mRNA codons

pp. 621–622

Reading a table of mRNA codons

D1.2.9



Amino acids by mRNA codon, read from the centre outwards. Mouagip, public domain.

mRNA	AUG | GCU | UGU | UAA
Polypeptide	Met – Ala – Cys – stop

Split the sequence into codons from the **first base**. Look up each in turn. A stop codon adds no amino acid.

Try it

Decode C A C G G G. Then write the DNA template strand for the same mRNA.

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Split the sequence into codons, then look up each in turn. A U G G C U U G U U A A gives Met, Ala, Cys, then stop. A stop codon adds no amino acid.

D1.2.10 How a ribosome builds a polypeptide

p. 622

A tRNA binds to the next codon. A **peptide bond** links the growing chain to the new amino acid. The ribosome moves one codon along, and the empty tRNA leaves. The cycle then repeats, adding one amino acid each time.

D1.2.11 Mutations that change protein structure

pp. 623–624

In **sickle cell anaemia**, one base in the gene for beta-globin is substituted. The codon GAG becomes GUG, so glutamic acid is replaced by valine. At low oxygen, haemoglobin forms long fibres, and red blood cells become rigid and sickle-shaped. They can block small blood vessels, and their short life leads to anaemia.

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